

# Jio Subscriber Retention & AI Chatbot Project

Day 1 Completion Report | Data Analysis & Diagnostics | Project-5 | Submitted 18 Sep 2026

## Scope

Per the homework brief, Day 1 covers three deliverables: (1) identify problematic regions, (2) analyse churn trends and timelines, and (3) check complaint correlations and important customer patterns. Analysis was run against the synthetic Jio Retention dataset (*Jio\_Retention\_Dataset.xlsx / subscribers.csv*, 64,738 subscriber records, calibrated to the engagement brief *Jio\_Retention\_Brief.pdf*), using Python (pandas / matplotlib / seaborn) in VS Code.

Dataset base rates: **1.73% churn (30-day)**, **5.01% churn (90-day)** across 64,738 subscribers. Three columns - `mnp_enquiry_flag`, `churn_reason`, `churn_date` - were excluded from all driver analysis below because they leak the churn outcome directly.

## A. Problematic Regions

Ranked circles by monthly churn % for the latest reported month (Aug 2026), cross-checked against each circle's own FY27 target and review status, and separately by port-out request volume.

### Top 6 circles by monthly churn %

| Circle                        | Churn % | ARPU (Rs) | Port-outs | Owner      | Status    |
|-------------------------------|---------|-----------|-----------|------------|-----------|
| Uttar Pradesh East            | 2.50    | 227.4     | 860,037   | N. Bose    | Watch     |
| Uttar Pradesh West            | 2.43    | 229.8     | 465,039   | R. Iyer    | Watch     |
| Bihar & Jharkhand             | 2.35    | 232.3     | 677,182   | R. Iyer    | On track  |
| Punjab                        | 2.33    | 230.4     | 173,557   | N. Bose    | On track  |
| Madhya Pradesh & Chhattisgarh | 2.24    | 226.5     | 590,414   | T. Nair    | Watch     |
| North East                    | 2.13    | 227.9     | 71,131    | V. Rathore | Escalated |

**Note on FY27 ceiling:** all 22 circles currently sit above their own FY27 churn ceiling, because the ceiling is a forward-looking reduction target set below today's actual for every circle - so "over ceiling" alone is not a discriminating filter. The more actionable signal is **review status**: circles the business has already flagged **Escalated** are **North East (2.13%)**, **Rajasthan (2.06%)**, **Haryana (1.82%)** and **Mumbai (1.32%)**. These combine both high absolute churn (North East, Rajasthan) and circles where churn is comparatively low but owner/budget attention is already stretched (Mumbai).

This is consistent with the engagement brief's claim that UP-East, UP-West, Madhya Pradesh and Bihar carry a disproportionate share of national port-out volume - all four rank in the top 5 on both churn % and absolute port-out requests.

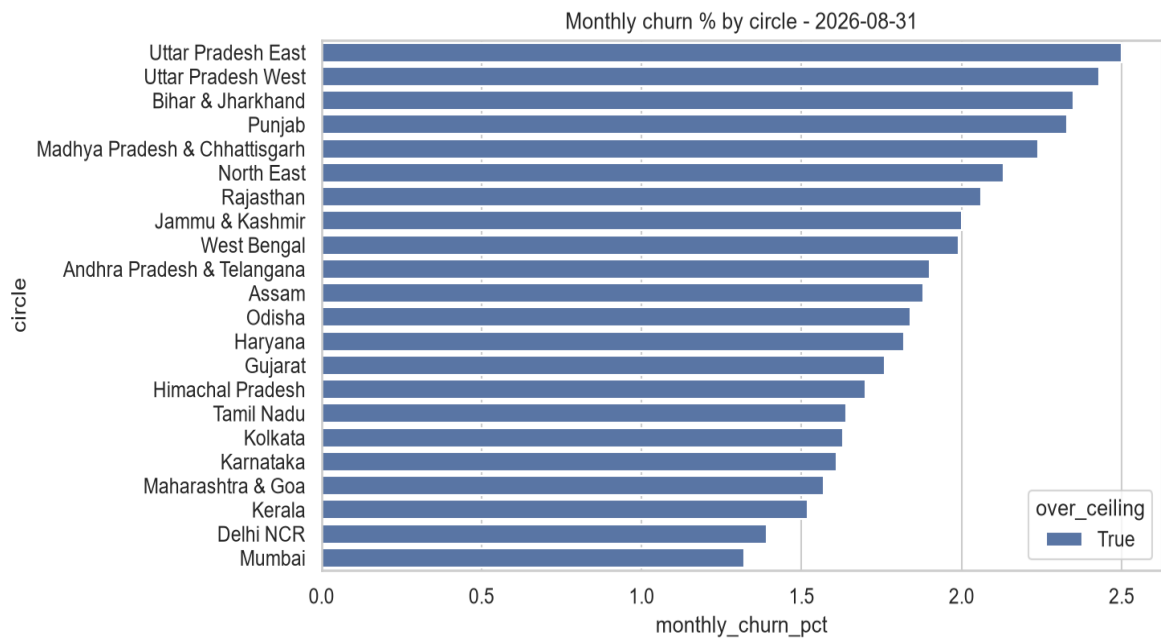


Chart 1 - Monthly churn % by circle, Aug 2026 (red = above FY27 ceiling)

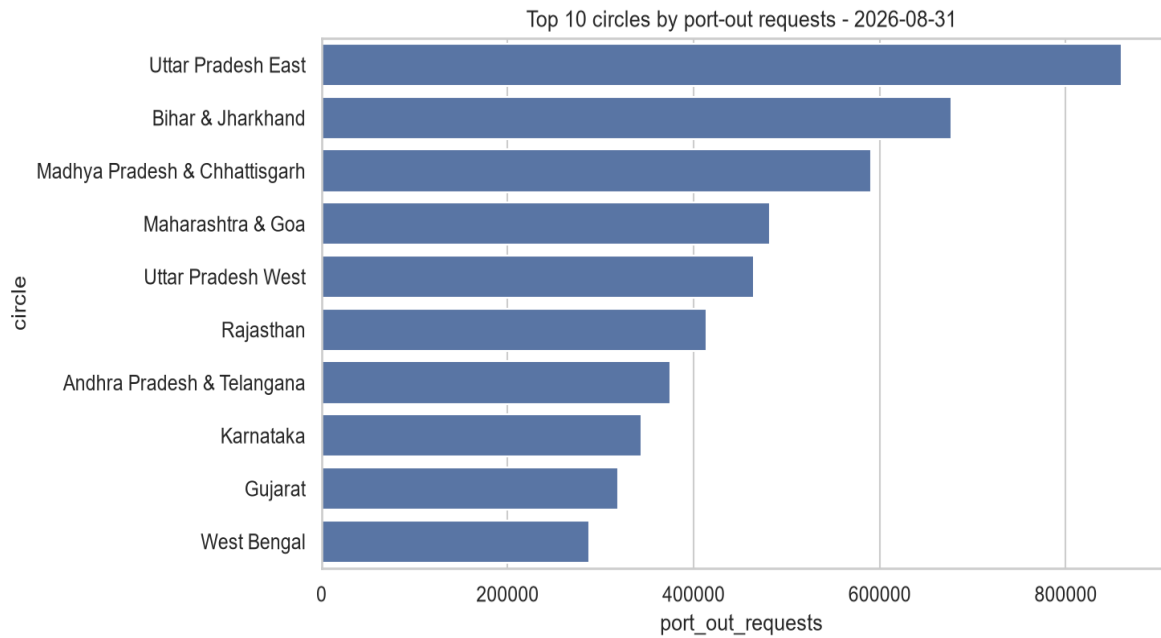
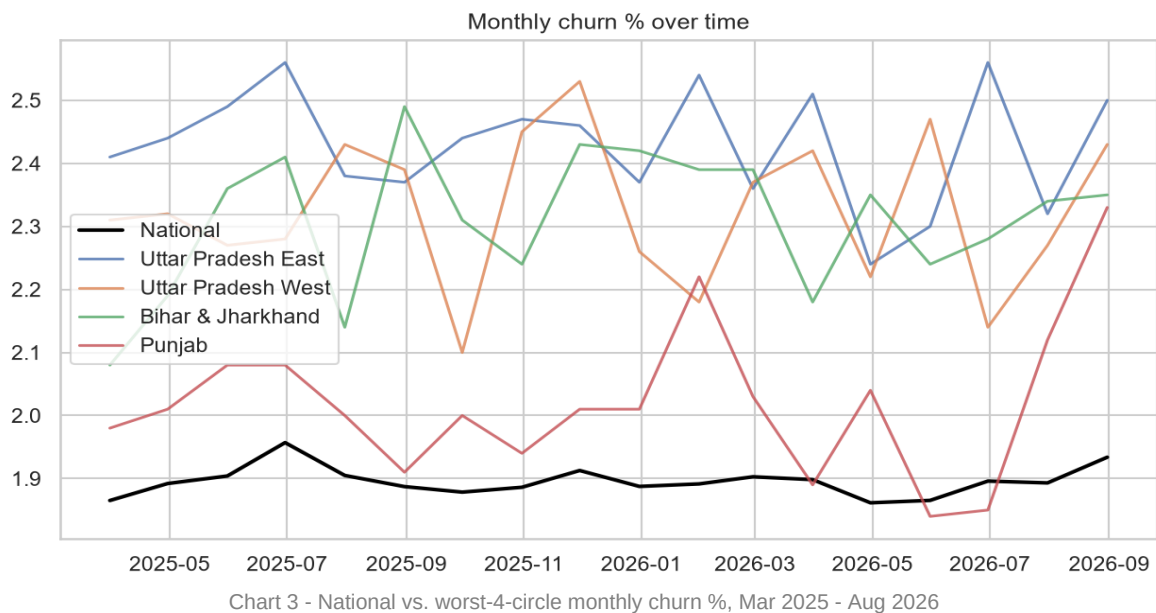


Chart 2 - Top 10 circles by port-out request volume, Aug 2026

## B. Churn Trends and Timelines

National base-weighted monthly churn has drifted from **1.87%** (Mar 2025) to **1.93%** (Aug 2026) - essentially flat with mild upward drift, matching the brief's description of churn "holding" with no sign of compression rather than a sudden spike.



### Tenure cliff

| Tenure bucket | Subscribers | 90-day churn rate |
|---------------|-------------|-------------------|
| 0-3m          | 568         | 14.44%            |
| 3-6m          | 2,066       | 10.45%            |
| 6-12m         | 7,021       | 7.26%             |
| 12-24m        | 17,345      | 5.79%             |
| 24-48m        | 24,281      | 4.23%             |
| 48m+          | 13,457      | 2.99%             |

There is a clear early-tenure cliff: subscribers in their **first 3 months churn at nearly 5x the rate of subscribers past 48 months** (14.44% vs 2.99% at 90 days), and risk decays smoothly and monotonically with tenure thereafter. This supports treating early-lifecycle onboarding as a distinct risk window in Day 2 modelling and in any retention offer design.

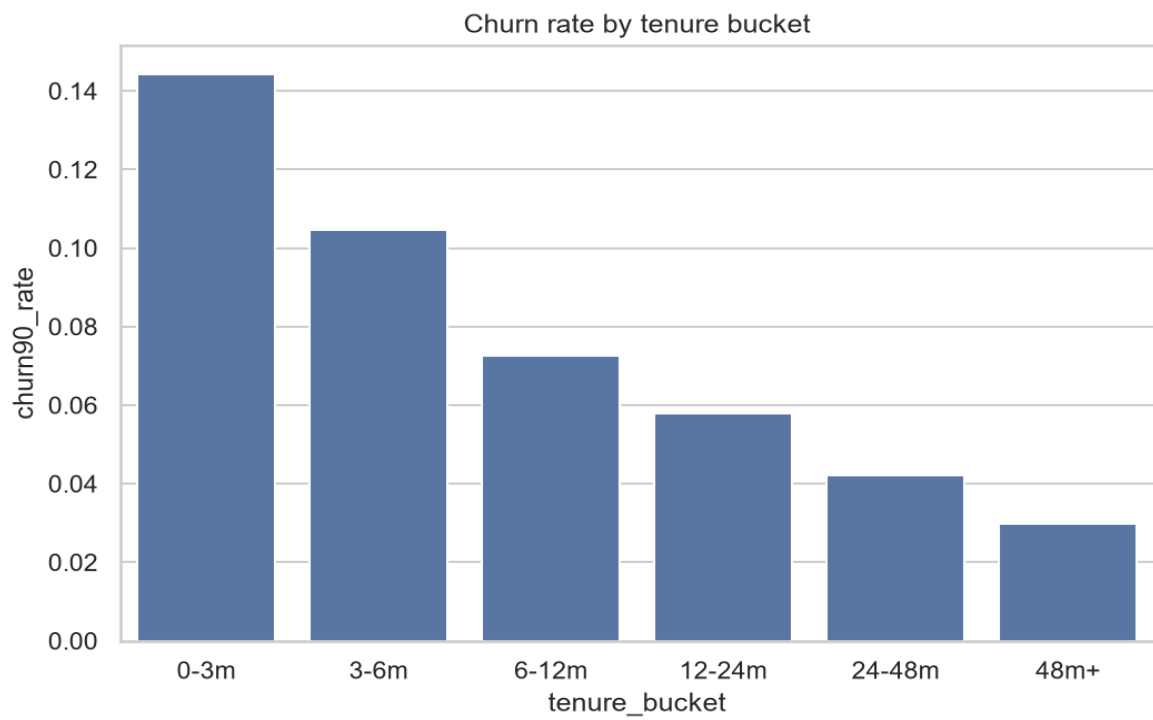
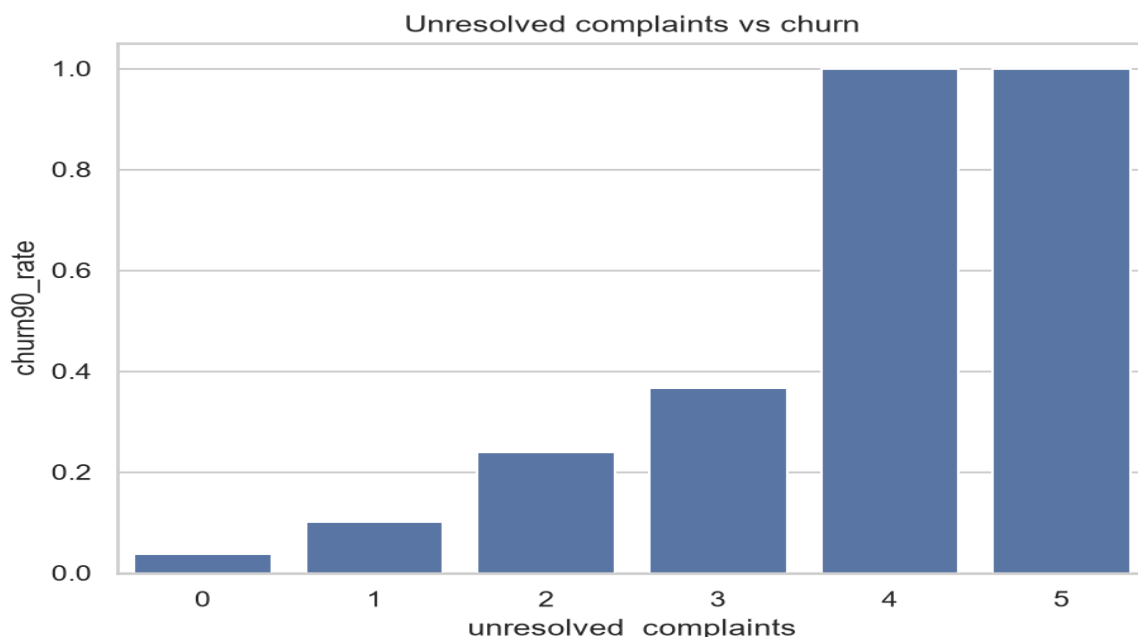


Chart 4 - 90-day churn rate by tenure bucket

## C. Complaint Correlations and Customer Patterns

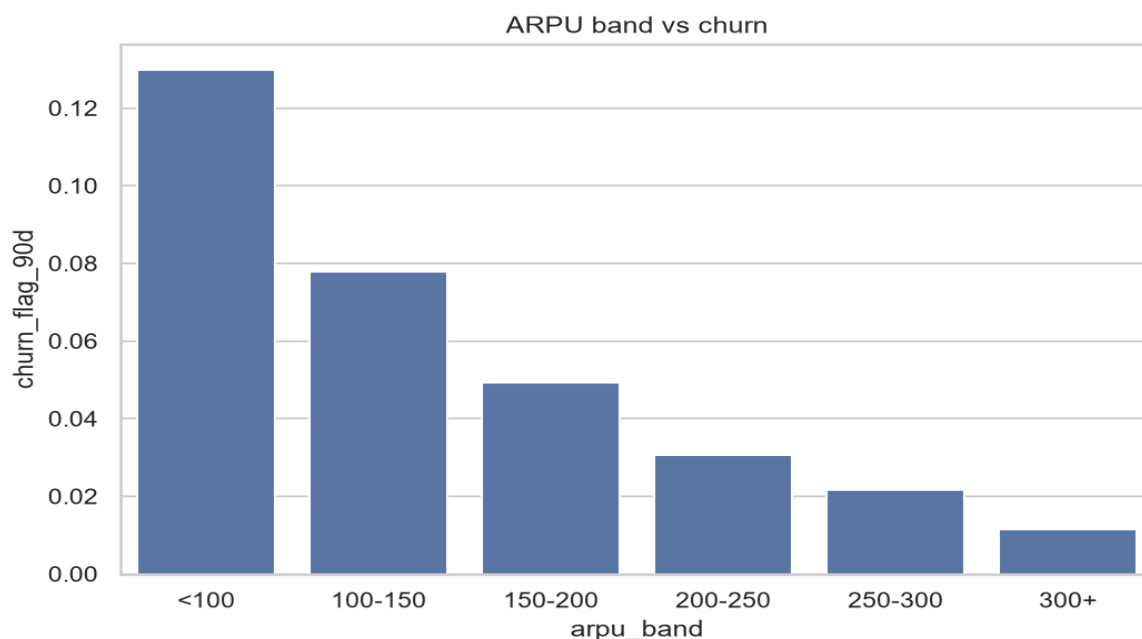
Four cuts of the data below: unresolved complaints, ARPU band, network signal quality, and an overall correlation ranking across every numeric feature. Each chart is explained in turn.

**Chart 5 - Unresolved complaints (6m) vs. 90-day churn rate**



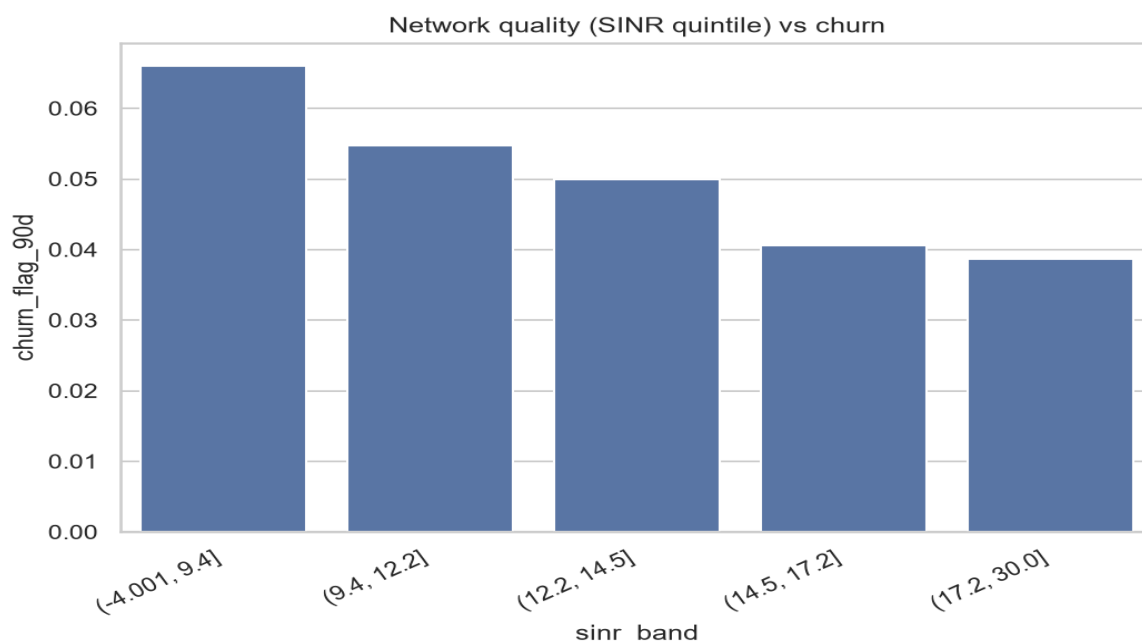
This chart buckets subscribers by how many complaints raised in the last 6 months are still unresolved, and plots the 90-day churn rate for each bucket. Churn climbs sharply and monotonically: **3.81%** at zero unresolved complaints, **10.09%** at one, **23.99%** at two, and **36.73%** at three. The 4-and-5 buckets show 100% churn, but each holds only a single subscriber, so that reading isn't statistically meaningful - the trend through 3 complaints is the reliable signal. The near-linear climb suggests each additional unresolved complaint independently compounds churn risk, rather than one issue tipping a subscriber over a threshold. This makes unresolved-complaint count one of the most operationally actionable levers available: closing tickets faster - even before the underlying issue is fixed - should reduce churn on its own.

**Chart 6 - ARPU band vs. 90-day churn rate**



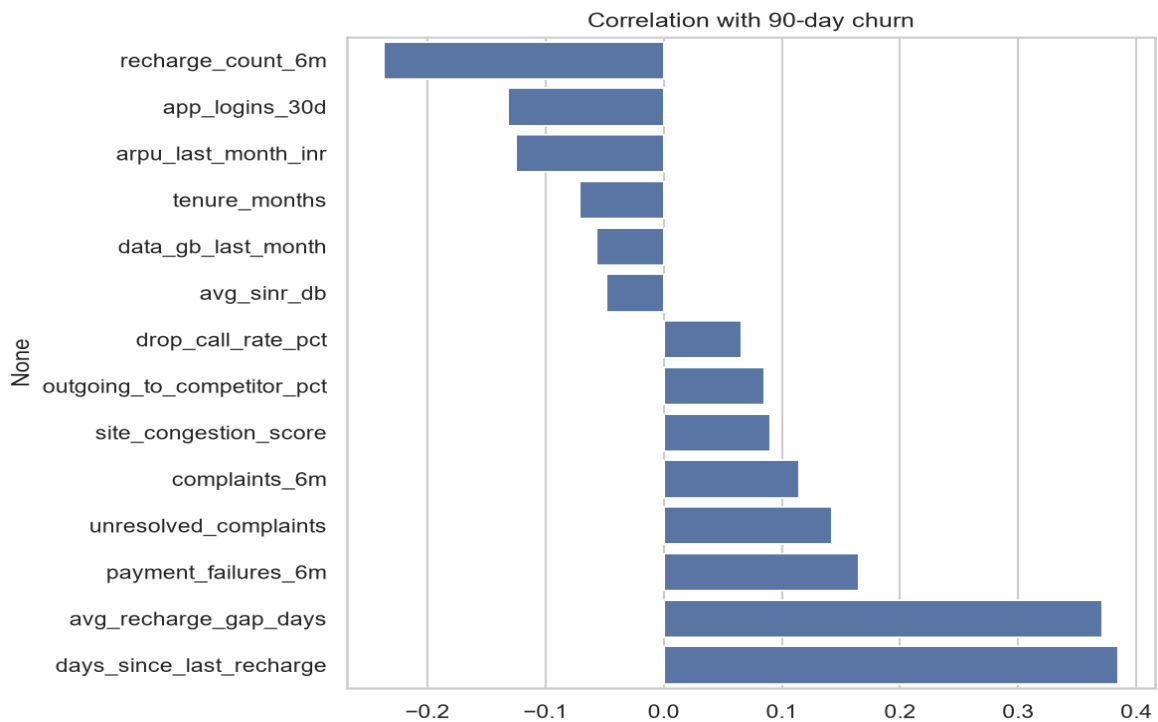
This chart splits subscribers into six ARPU bands, from under Rs.100 up to Rs.300+, and plots the 90-day churn rate in each band. Churn falls steadily from **12.99%** in the lowest band to **1.15%** in the highest - an **11x spread** across the ARPU range. The decline is smooth and monotonic with no plateau, meaning every step up in ARPU reduces churn risk rather than only the extreme bands mattering. This directly confirms the engagement brief's claim that the low-ARPU tail is the most price-elastic, highest-churn segment of the base. For Day 2, ARPU band is a natural segmentation axis for both the churn model and for targeting retention offers where they will have the largest effect.

### Chart 7 - Network quality (SINR quintile) vs. 90-day churn rate



This chart splits subscribers into five quintiles by average SINR (signal quality), from weakest signal on the left to strongest on the right, and plots 90-day churn per quintile. Churn is highest in the weakest-signal quintile (**~6.6%**) and lowest in the strongest-signal quintile (**~3.8%**). The drop across quintiles is real but far shallower than the ARPU or complaints effect - roughly a **1.7x spread**, versus 11x for ARPU. This suggests network quality is a secondary, contributing churn factor in this dataset rather than a primary driver. It is still worth carrying into the Day 2 feature set, but should not be prioritized over recharge behaviour, complaints, or ARPU when deciding where to act first.

### Chart 8 - Correlation of numeric features with 90-day churn



This chart ranks every numeric subscriber attribute by its Pearson correlation with 90-day churn, from strongest protective factor at top to strongest risk driver at bottom. **recharge\_count\_6m** is the strongest protective factor (**-0.237**) - subscribers who recharge more often are markedly less likely to churn.

**days\_since\_last\_recharge** (**+0.385**) and **avg\_recharge\_gap\_days** (**+0.371**) are by far the strongest risk drivers, well ahead of any complaint or network metric. Complaint and network features (**unresolved\_complaints** +0.143, **drop\_call\_rate\_pct** +0.066, **avg\_sinr\_db** -0.048) sit in the middle of the ranking - real effects, but secondary to recharge behaviour. This ranking is the direct input to Day 2 feature selection: the top 4-5 features by absolute correlation should form the backbone of the 30-day churn model.

### 5G adoption (supporting context)

5G-active subscribers churn at **3.42%** versus **6.09%** for non-5G-active (26,265 vs 38,473 subscribers) - consistent with 5G adoption acting as an engagement/stickiness signal rather than a churn cause in its own right.

## Key Takeaways

- Regional focus for phase 1 pilots: North East, Rajasthan, Haryana and Mumbai (Escalated status), plus UP-East, UP-West, MP and Bihar (highest absolute churn/port-out volume).
- Onboarding is the single highest-risk lifecycle window - first 3 months churn at ~5x the rate of 48m+ tenure subscribers.
- Recharge behaviour (recency and gap) is the strongest available early-warning signal, ahead of complaints and network quality - a strong candidate as the top feature for the Day 2 churn model.
- Unresolved complaints and low ARPU both show large, monotonic churn effects and are natural segments for targeted retention offers.
- 5G adoption is protective, not causal - useful as a model feature but not a lever to pull directly.

## Next Steps (Day 2 Preview)

Feature set for the 30-day churn model, CLV regression and offer-response/uplift model will draw primarily on recharge recency/gap, payment failures, unresolved complaints, ARPU and tenure - the features ranked highest here - after excluding the three leakage columns identified on Day 1.

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Sources: outputs/day1\_\*.csv, outputs/charts/\*.png, generated from data/raw/subscribers.csv and data/raw/Jio\_Retention\_Dataset.xlsx via scripts/day1\_data\_analysis.py.